

MOHAMMAD QATAADAH ZAKIR

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer specializing in production-ready AI applications, LLM systems, retrieval-augmented generation (RAG), and scalable backend infrastructure. Experienced designing and deploying AI applications using Python, FastAPI, Docker, AWS, and vector databases, with a focus on deploying reliable, high performance AI software solutions.

WORK EXPERIENCE

AI Software Engineer | Optimal Communication Co.

July 2025 – Present

- Engineered and deployed AI-powered payment automation solutions for POS terminals, card payment systems, and internal business workflows, developing scalable backend services with Python, FastAPI, REST APIs, SQL, and Git that reduced manual processing by 65%, improved transaction accuracy by 30%, and supported 15+ production features.
- Designed, built, and maintained full-stack software applications from requirements through deployment, integrating payment gateways, third-party APIs, authentication systems, databases, and business automation tools, decreasing workflow completion by 45% while improving system reliability and maintainability through comprehensive testing, debugging, and code reviews.
- Collaborated closely with product managers, QA engineers, and stakeholders to deliver 10+ software enhancements and automation initiatives, optimizing application performance, resolving critical production issues, and implementing CI/CD and version control best practices that reduced deployment times by 40% and improved release stability.

Software Engineer – Intern | Modern Arc Group

May 2024 – August 2024

- Assisted in the development and enhancement of web-based business applications using modern software engineering practices, contributing to frontend and backend development, REST API integration, SQL database management, debugging, testing, and documentation, while helping deliver 5+ client and internal software solutions within project deadlines.
- Worked alongside senior software engineers to analyze business requirements, implement new features, troubleshoot complex software defects, and participate in peer code reviews, successfully resolving 70+ reported issues and improving application performance, maintainability, and user experience across multiple development cycles.
- Contributed to software deployment, database optimization, and technical documentation while gaining experience with Git, Agile/Scrum methodologies, SDLC, version control, and cross-functional collaboration, reducing onboarding time for new developers by 30% and supporting faster feature delivery throughout the internship.

Freelance Software Engineer

June 2022 – January 2024

EDUCATION

Software Engineering, B.Eng. (Hons)

University of Greater Manchester (formerly University of Bolton) | 2025

Computer Engineering, B.Eng.

Memorial University of Newfoundland (MUN) | 2022 (Credit transfer)

PROJECTS

LLM Fine-Tuning & Inference Pipeline

Designed and deployed a production-ready large language model (LLM) training and inference pipeline, covering parameter-efficient fine-tuning, model optimization, evaluation, and high-performance deployment. The project demonstrates expertise in modern LLM engineering, scalable inference, and efficient deployment on consumer and cloud hardware. It incorporates LoRA/QLoRA fine-tuning, quantization, Hugging Face Transformers, and llama.cpp to optimize model performance while reducing computational requirements for real-world deployment.

Enterprise Retrieval-Augmented Generation (RAG) System

Built a production-grade Retrieval-Augmented Generation (RAG) platform for intelligent document search and question answering, integrating modern retrieval techniques, LLM inference, and scalable backend infrastructure. The project emphasizes low-latency retrieval, high-quality contextual responses, and cloud-ready deployment. Leveraging vector databases, semantic embeddings, and FastAPI, the system delivers context-aware responses across large document collections while minimizing hallucinations and maintaining fast query performance.

QEMI - AI Development Platform

Developed an AI-powered software engineering platform that combines real-time system monitoring, runtime diagnostics, and large language models to assist developers with intelligent debugging and code analysis. The platform was designed as a production-ready MVP and demonstrated to potential clients for early-stage commercial adoption. Built with a modern React frontend and Python-based backend, QEMI integrates AI-assisted diagnostics, automated error analysis, and developer tooling to streamline debugging workflows and improve software productivity.

SKILLS

Programming languages – Python, SQL, Typescript, Bash,

Frameworks & Libraries – PyTorch, Tensorflow, Scikit-learn, XGBoost, Hugging Face, llama-cpp, ONNX

LLMs & AI Frameworks – OpenAI API, Claude, Cursor, Embedding APIs, Vector Databases, LangChain, Together AI, Zapier, LangGraph

Cloud, Data & Infrastructure – AWS (SageMaker, EC2, S3), Azure, Docker, Kubernetes, PostgreSQL, MongoDB, Supabase, Git/Github, REST

Soft skills – Agile, Cross-functional team, Communication, Client Collaboration, Documentation, Adaptability, Time Management, Emotional Intelligence

CERTIFICATIONS

- EDUCBA – AI with Python: Foundations to Projects Specialization
- DeepLearnin.AI – Deep Learning Specialization
- Duke University – MLOps | Machine Learning Operations Specialization
- Stanford Online – Machine Learning Specialization
- HarvardX - Computer Science for Python Programming
- HarvardX - Computer Science for Artificial Intelligence
- IBM – AI Chatbots without Programming
- IBM – Python basics for Data Science
- IBM – AI Applications with Watson
- University of Minnesota – Software Development Lifecycle